Course Code: SBT-DF204

Lab Title: Case Study 4: Investigating Rhion Possession Evidence

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Executive Summary

This investigation examined a USB image, three network-capture files, reconstructed JPEG images, an encrypted ZIP archive, and HTTP objects exported from the primary capture. The USB image was identified as FAT16 and contained two allocated text files containing recipes. Network analysis of rhino.log identified FTP uploads named rhino1.jpg, rhino3.jpg, and contraband.zip.

The reconstructed JPEG images were validated, and the two reconstructed versions of rhino3.jpg had matching SHA-256 hashes. The ZIP archive was identified as encrypted, but its image contents were not recovered. Basic metadata, file-signature, and empty-passphrase steganography checks did not confirm hidden content.

The findings document file handling and network transfers. However, they do not independently prove who controlled the files, knew their contents, or possessed contraband.

Lab Objective

The objective of this investigation was to examine the supplied USB image and network captures, recover and validate available files, check for possible hidden or protected content, establish a timeline of relevant events, and document the findings using digital forensic methods.

The investigation also aimed to distinguish direct technical observations from interpretations about possession, file ownership, and user attribution.

Tools and Resources Used

The investigation was performed in Kali Linux from the working directory /home/kali/rhion.

The tools used included:

- The Sleuth Kit 4.12.1
- TShark 4.6.6
- Foremost 1.5.7

- ExifTool 13.25
- Steghide 0.6.0
- Binwalk

Evidence Preservation and Integrity

The working directory contained the USB image, three network-capture files, reconstructed images, a reconstructed ZIP archive, selected TCP stream exports, and an HTTP object export directory.

SHA-256 hashes were recorded for the principal evidence files and reconstructed objects. These values can be used to verify that the files have not changed since the hashes were calculated.

SHA-256 Hashes

Evidence item	**SHA-256**
RHINOUSB.dd	ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65
rhino.log	64e6d55b76660eb3aaa41572c1d04e4452510a343bbf42e844424827dedfddb2
rhino2.log	41939d5de0556b70279056572dee44b6fd84cd05b0788cfd0b6f52f37b161dde
rhino3.log	7b0304f5e88a30c305a99b5a1e2977bced5b9c94da6f458f887bb39966bbfc46
rhino1_recovered.jpg	0f30a8777d19a4fe4b2fafd5154b6c4dfafb673202db29971e876e978da4e05d
rhino3_recovered.jpg	a85aea8e47d9f5425456f3bafb9c67201c8c2b2b2cec80154c83102e802240c5
rhino3_binary_recovered.jpg	a85aea8e47d9f5425456f3bafb9c67201c8c2b2b2cec80154c83102e802240c5

contraband_recovered.zip	b7091a0b4295a561bf76969073ede126a5ca75899bb115447e67272f95d5a29f

Analysis and Findings

USB Image Examination

The image RHINOUSB.dd was identified as a FAT16 filesystem. The filesystem report showed no sectors preceding the filesystem, so the image was examined directly without applying a partition offset.

The filesystem details included:

- Filesystem: FAT16
- Volume identifier: 0x4092d9d1
- Sector size: 512 bytes
- Cluster size: 4096 bytes

The root directory listing contained two allocated files:

- gumbo1.txt, metadata record 4
- gumbo2.txt, metadata record 6

Both files contained recipe text. The deleted-entry search using fls -r -d returned no entries. This means that the command did not list deleted entries; it does not prove that no deleted or recoverable data exists anywhere in the image.

The metadata for gumbo1.txt recorded a size of 2,815 bytes and a written/created timestamp of 30 April 2004 at 18:11:20 EDT. The metadata for gumbo2.txt recorded a size of 1,293 bytes and a timestamp of 18:11:24 EDT.

Converted to UTC, the timestamps are 30 April 2004 at 22:11:20Z and 22:11:24Z respectively. These are filesystem timestamps and should not be treated as independently verified file-copy times.

FTP Traffic Analysis

The primary network capture, rhino.log, contained FTP and FTP-data traffic.

The FTP client was 137.30.122.253, and the server was 137.30.120.40.

The following FTP upload commands were identified:

Frame	Filename	Capture time (UTC)	Data stream
1546	rhino1.jpg	26 April 2004, 22:21:49.954047Z	71
1649	rhino3.jpg — ASCII mode	26 April 2004, 22:22:08.905569Z	79
1763	rhino3.jpg — binary mode	26 April 2004, 22:22:16.091224Z	83
5647	contraband.zip	26 April 2004, 22:26:46.671569Z	307

The timestamps were converted from EDT (UTC−04:00) to UTC by adding four hours.

Reconstructed JPEG Images

The file rhino1_recovered.jpg was identified as a valid JPEG image using JFIF version 1.02, with dimensions of 501 × 360 pixels.

The files rhino3_recovered.jpg and rhino3_binary_recovered.jpg were identified as valid JPEG images using JFIF version 1.01, with dimensions of 720 × 597 pixels.

Both reconstructed versions of rhino3.jpg had the same SHA-256 hash:

a85aea8e47d9f5425456f3bafb9c67201c8c2b2b2cec80154c83102e802240c5

This supports that the two recovered versions are byte-identical. The ASCII-mode transfer may have altered file bytes, so the binary-mode reconstruction provides a useful comparison.

Encrypted ZIP Archive

The reconstructed archive contraband_recovered.zip was identified as a ZIP archive. Its listing showed one member named rhino2.jpg, with an uncompressed size of 230,665 bytes.

Archive inspection indicated that the member was encrypted. An extraction attempt was cancelled at the password prompt, so the contents of rhino2.jpg were not examined.

The SHA-256 hash of the reconstructed archive was:

b7091a0b4295a561bf76969073ede126a5ca75899bb115447e67272f95d5a29f

Metadata and Hidden-Content Checks

ExifTool was used to review metadata in the reconstructed JPEG images.

The file rhino1_recovered.jpg contained a Photoshop thumbnail and a Photoshop 4.0 comment. Binwalk detected JPEG signatures at offsets 0 and 530. The second signature may correspond to the thumbnail indicated by the metadata; it is not, by itself, proof of steganography.

For rhino3_recovered.jpg, Binwalk detected a JPEG signature at offset 0 only.

Steghide was tested on both images using an empty passphrase. Neither test extracted data.

HTTP Traffic and Exported Objects

HTTP objects were exported from rhino.log using TShark. A subsequent count of the files in the http_objects directory returned 375 objects.

The sampled HTTP responses included HTML, JavaScript, CSS, GIF, and JPEG content, including webmail resources. The count alone does not identify which objects are relevant to the investigation. The sampled output did not establish a relevant HTTP-delivered archive or executable.

A case-related email titled “New rhino pics” was observed in TCP stream 259. The message referred to planned uploads of rhino material. This provides context for the network activity, but it does not independently establish the identity or intent of the person operating the account.

Timeline of Events (UTC)

The network capture timestamps were displayed in EDT (UTC−04:00) and converted to UTC. The USB timestamps are filesystem metadata and are listed separately because they do not independently establish file-copy events.

Date and time (UTC)	Events	Source or reference
26 April 2004, 22:18:40.957966Z	Start of rhino.log capture	Capture timestamp
26 April 2004, 22:21:49.954047Z	FTP upload of rhino1.jpg	Frame 1546; data stream 71
26 April 2004, 22:22:08.905569Z	FTP upload of rhino3.jpg in ASCII mode	Frame 1649; data stream 79
26 April 2004, 22:22:16.091224Z	FTP upload of rhino3.jpg in binary mode	Frame 1763; data stream 83
26 April 2004, 22:26:46.671569Z	FTP upload of contraband.zip	Frame 5647; data stream 307
30 April 2004, 22:11:20Z	Filesystem timestamp for gumbo1.txt	istat record 4
30 April 2004, 22:11:24Z	Filesystem timestamp for gumbo2.txt	istat record 6

The recorded capture ranges were:

- rhino.log: 26 April 2004, 18:18:40.957966 to 18:31:11.151543 EDT.
- rhino2.log: 28 April 2004, 17:08:30.180553 to 17:10:09.295803 EDT.
- rhino3.log: 28 April 2004, 17:15:41.191431 to 17:16:48.642060 EDT.

Evidence Log

Evidence ID	Source and method	Finding	Screenshot reference
E01	SHA-256 and tool-version checks	Integrity values recorded for the USB image, PCAPs, recovered JPEGs, and ZIP archive	Screenshot 1,2
E02	The Sleuth Kit: fsstat, fls, istat, and icat	FAT16 image; two allocated recipe files; no deleted entries listed by the deleted-entry search	Screenshot 3
E03	TShark FTP filters and data streams	Uploads named rhino1.jpg, rhino3.jpg, and contraband.zip; timestamps and stream IDs recorded	Screenshot 4,5,6
E04	TCP stream 307 reconstruction and ZIP inspection	Encrypted ZIP archive; rhino2.jpg listed but not decrypted	Screenshot 9
E05	File identification, ExifTool, Binwalk, and Steghide	JPEGs validated; no data extracted with an empty passphrase; second JPEG signature in rhino1_recovered.jpg may be a thumbnail	Screenshot 7,8,10,11,12
E06	TShark HTTP object export and response review	The http_objects directory contained 375 files at the time of the check. Sampled responses included webmail resources and other web content.	Screenshot 13

Conclusion

The examination established that the USB image contains a FAT16 filesystem with two allocated recipe text files. FTP traffic in rhino.log documents uploads named rhino1.jpg, rhino3.jpg, and contraband.zip.

Two distinct JPEG images were recovered and examined. The two reconstructed versions of rhino3.jpg shared the same SHA-256 hash, confirming that their contents were identical. The ZIP archive was identified as encrypted, but its image member could not be inspected because the password was not recovered. Basic metadata, file-signature, and empty-passphrase steganography checks did not confirm hidden content in the examined JPEGs.

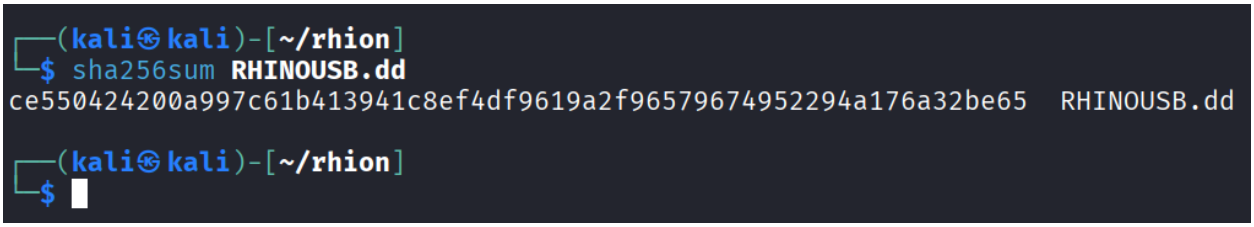
The email and FTP events provide contextual links between communications and file uploads. However, they do not independently prove that a specific person controlled the client, knew the contents of the transferred files, or possessed contraband.

The encrypted archive, incomplete review of HTTP objects, possible alternative steganography methods, and limitations of user attribution remain important limitations of the investigation.

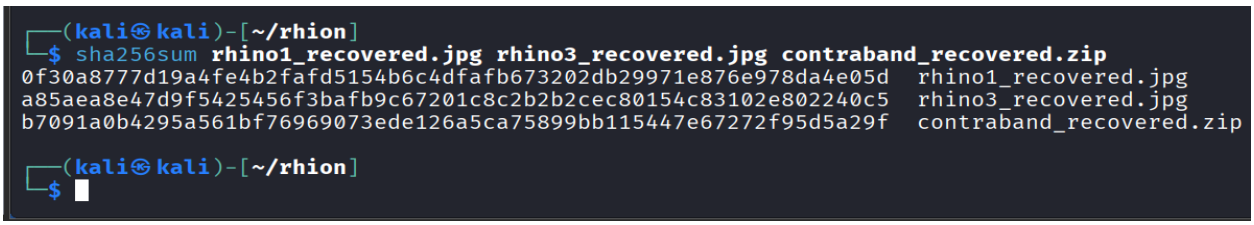
Appendix

The following screenshots support the analysis and findings presented in this report:

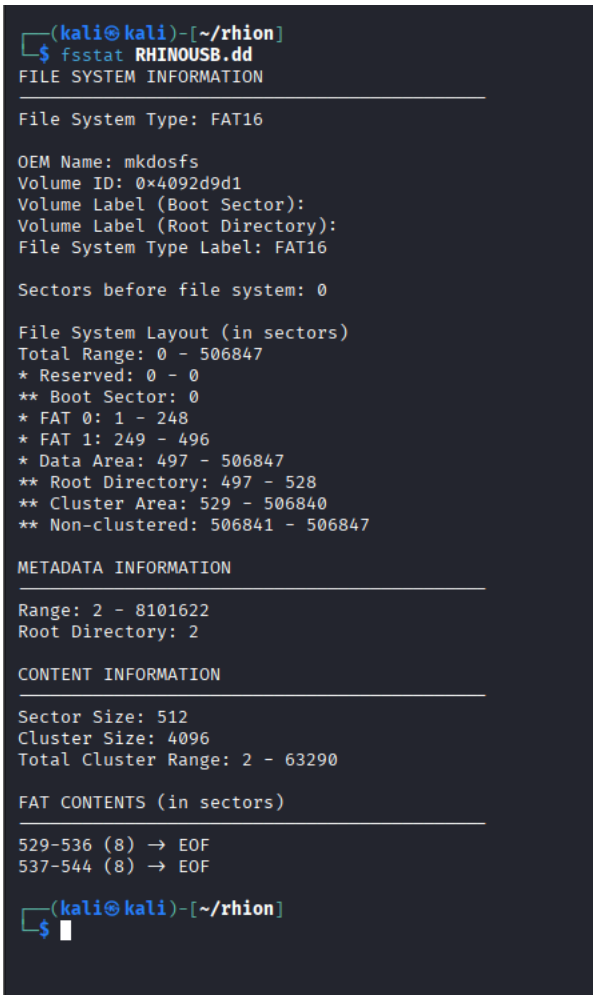
- SS01 — USB Image Integrity Hash



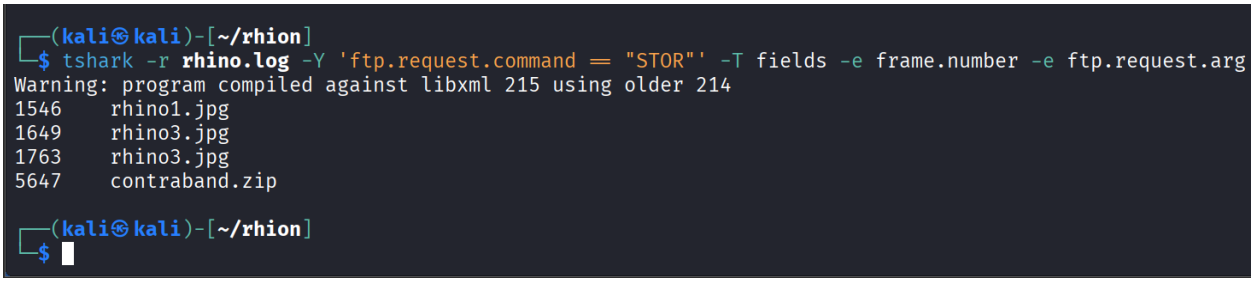
- SS02 — Recovered Evidence Hashes



- SS03 — USB Filesystem Analysis



- SS04 — FTP File Upload Evidence



- SS05 — FTP Upload Timeline

```
┌──(kali㉿kali)-[~/rhion]
└─$ tshark -r rhino.log -Y 'ftp.request.command == "STOR"' -T fields -e frame.number -e frame.time -e ftp.request.arg
Warning: program compiled against libxml 215 using older 214
1546    2004-04-26T18:21:49.954047000-0400      rhino1.jpg
1649    2004-04-26T18:22:08.905569000-0400      rhino3.jpg
1763    2004-04-26T18:22:16.091224000-0400      rhino3.jpg
5647    2004-04-26T18:26:46.671569000-0400      contraband.zip

┌──(kali㉿kali)-[~/rhion]
└─$
```

- SS06 — FTP Credentials in Network Capture (redacted)

```
┌──(kali㉿kali)-[~/rhion]
└─$ tshark -r rhino.log -Y 'ftp.request.command == "USER" || ftp.request.command == "PASS"' -T fields -e frame.number -e
 ftp.request.command -e ftp.request.arg
Warning: program compiled against libxml 215 using older 214
1532    USER    gnome
1536    PASS    gnome123
1625    USER    gnome
1629    PASS    gnome123
5633    USER    gnome
5637    PASS    gnome123

┌──(kali㉿kali)-[~/rhion]
└─$
```

- SS07 — RHINO3 Image Integrity Check

```
┌──(kali㉿kali)-[~/rhion]
└─$ sha256sum rhino3_recovered.jpg rhino3_binary_recovered.jpg
a85aea8e47d9f5425456f3bafb9c67201c8c2b2b2cec80154c83102e802240c5  rhino3_recovered.jpg
a85aea8e47d9f5425456f3bafb9c67201c8c2b2b2cec80154c83102e802240c5  rhino3_binary_recovered.jpg

┌──(kali㉿kali)-[~/rhion]
└─$
```

- SS08 — Recovered Image File Details

```
┌──(kali㉿kali)-[~/rhion]
└─$ file rhino1_recovered.jpg rhino3_recovered.jpg
rhino1_recovered.jpg: JPEG image data, JFIF standard 1.02, resolution (DPI), density 72×72, segment length 16, comment:
"File written by Adobe Photoshop\250 4.0", baseline, precision 8, 501×360, components 3
rhino3_recovered.jpg: JPEG image data, JFIF standard 1.01, resolution (DPI), density 72×72, segment length 16, baseline,
 precision 8, 720×597, components 3

┌──(kali㉿kali)-[~/rhion]
└─$
```

- SS09 — Encrypted ZIP Contents

```
┌──(kali㉿kali)-[~/rhion]
└─$ unzip -l contraband_recovered.zip
Archive:  contraband_recovered.zip
  Length      Date    Time    Name
---------  ---------- -----   ----
   230665  2004-04-26 17:00   rhino2.jpg
---------                     -------
   230665                     1 file

┌──(kali㉿kali)-[~/rhion]
└─$
```

- SS10 — JPEG Metadata Review (if saved)

```
┌──(kali㉿kali)-[~/rhion]
└─$ exiftool rhino1_recovered.jpg rhino3_recovered.jpg
======== rhino1_recovered.jpg
ExifTool Version Number         : 13.25
File Name                       : rhino1_recovered.jpg
Directory                       : .
File Size                       : 66 kB
File Modification Date/Time     : 2026:10:24 09:26:46-04:00
File Access Date/Time           : 2026:10:26 04:46:29-04:00
File Inode Change Date/Time     : 2026:10:24 09:26:46-04:00
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.02
Resolution Unit                 : inches
X Resolution                    : 72
Displayed Units X               : inches
Y Resolution                    : 72
Displayed Units Y               : inches
Copyright Flag                  : False
Photoshop BGR Thumbnail         : (Binary data 4999 bytes, use -b option to extract)
Photoshop Quality               : 7
Photoshop Format                : Standard
Comment                         : File written by Adobe Photoshop◆ 4.0
DCT Encode Version              : 100
APP14 Flags 0                   : (none)
APP14 Flags 1                   : (none)
Color Transform                 : YCbCr
Image Width                     : 501
Image Height                    : 360
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
```

```
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 501×360
Megapixels                      : 0.180
======== rhino3_recovered.jpg
ExifTool Version Number         : 13.25
File Name                       : rhino3_recovered.jpg
Directory                       : .
File Size                       : 97 kB
File Modification Date/Time     : 2026:10:24 09:29:26-04:00
File Access Date/Time           : 2026:10:26 04:46:29-04:00
File Inode Change Date/Time     : 2026:10:24 09:29:26-04:00
File Permissions                : -rw-rw-r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : inches
X Resolution                    : 72
Y Resolution                    : 72
Image Width                     : 720
Image Height                    : 597
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:2:0 (2 2)
Image Size                      : 720×597
Megapixels                      : 0.430
    2 image files read

┌──(kali㉿kali)-[~/rhion]
└─$
```

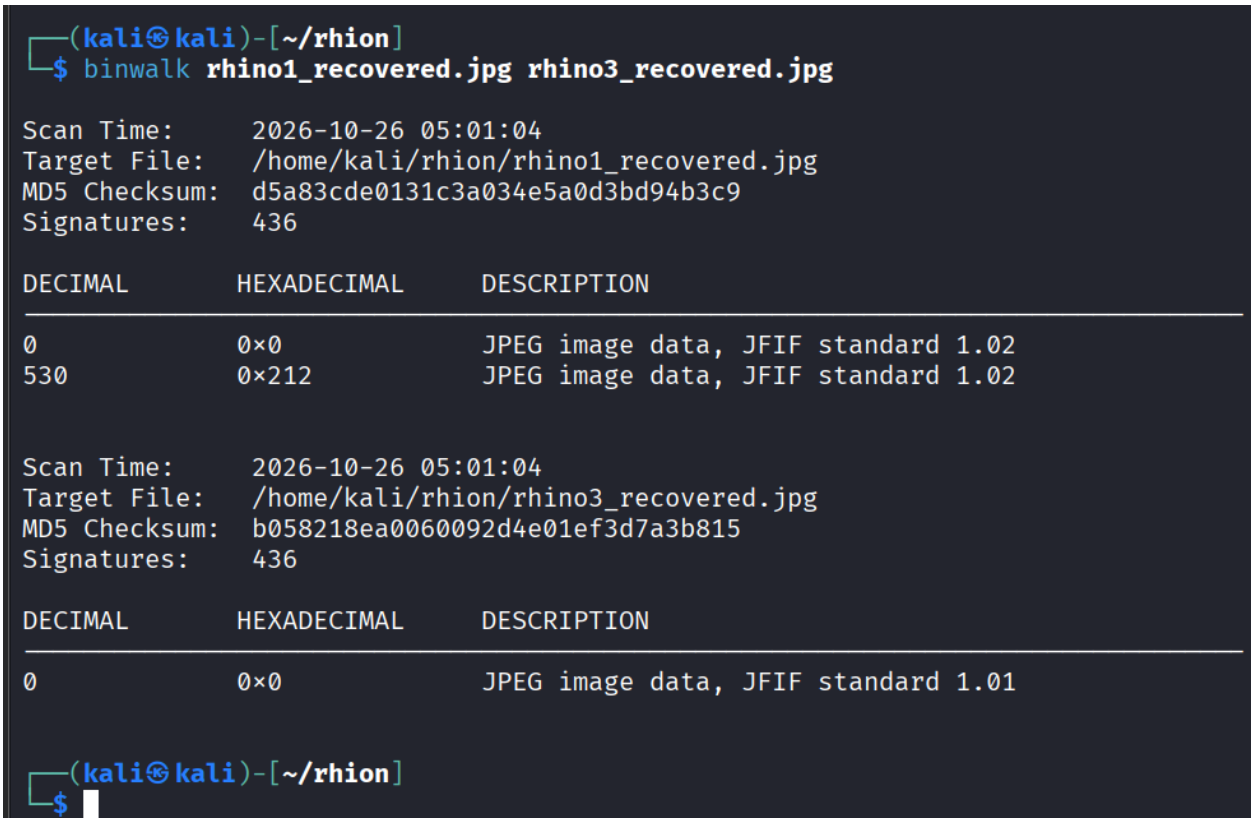
- SS11 — Steganography Check

```
┌──(kali㉿kali)-[~/rhion]
└─$ steghide info rhino1_recovered.jpg
"rhino1_recovered.jpg":
  format: jpeg
  capacity: 3.4 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
steghide: could not extract any data with that passphrase!

┌──(kali㉿kali)-[~/rhion]
└─$ steghide info rhino3_recovered.jpg
"rhino3_recovered.jpg":
  format: jpeg
  capacity: 5.4 KB
Try to get information about embedded data ? (y/n) y
Enter passphrase:
steghide: could not extract any data with that passphrase!

┌──(kali㉿kali)-[~/rhion]
└─$
```

- SS12 — Binwalk File-Signature Check



- SS13 — HTTP Object Extraction



Appendix B: Supporting Evidence Files

The investigation used the following principal evidence files:

- RHINOUSB.dd — USB forensic image.
- rhino.log — primary network capture containing FTP and HTTP traffic.
- rhino2.log — additional network capture.
- rhino3.log — additional network capture.
- rhino1_recovered.jpg — reconstructed JPEG image.
- rhino3_recovered.jpg — reconstructed JPEG image.
- rhino3_binary_recovered.jpg — binary-mode reconstruction of rhino3.jpg.
- contraband_recovered.zip — reconstructed encrypted ZIP archive.

The reconstructed ZIP archive remains encrypted, and its member rhino2.jpg was not recovered for inspection. The exported HTTP objects are stored in the http_objects directory.

Appendix C: Investigation Limitations

1. The encrypted ZIP archive could not be inspected because its password was not recovered.
2. The empty-passphrase steganography tests did not extract data, but other passphrases and hiding methods were not ruled out.
3. The HTTP object count records the current directory contents and does not independently confirm the origin or relevance of every file.
4. Network transfers and email content do not, by themselves, establish the identity of the person operating an account or their knowledge of a file's contents.
5. Filesystem timestamps may not independently establish when a file was copied or transferred.